SecureDT Data Flow and Implementation Guide

Intelligent and Secure Task Scheduling for Digital Twin-Enabled Edge-Cloud Networks

1. USER
User interacts with the system through a web browser. All communication goes through the Django backend. User actions: login (session-based authentication), register (new account creation), dashboard (overview of system state), dataset manager (browse, upload, import datasets), data upload (CSV file upload), data processing (initiate processing pipeline), AI module (run risk analysis), genetic algorithm (configure and run scheduling), digital twin (view resource state), task routing (view/reconfigure task assignments), security monitor (view security events), attack simulation (launch controlled attacks), storage attack (simulate storage threats), analytics (view results and charts), experiments (run comparison campaigns), reports (generate PDF/CSV/JSON reports), system health (check service status), settings (configure application parameters).

2. LOGIN
Purpose: Authenticate user and establish session. What user sees: Login form (username, password) or dashboard if already authenticated. What buttons do: Submit credentials, link to register password reset. Input: username, password. Output: Session cookie set, redirect to dashboard (or 401 if invalid). Expected result: Authenticated session; all protected endpoints require valid session.

3. DATASET
Purpose: Manage synthetic and real datasets for the system. What user sees: Dataset registry listing built-in datasets (tasks, attacks, resources, experiments, AI training, storage security), upload interface, big-data generation controls. What buttons do: Upload CSV, generate big-data (100K-10M+ rows), validate dataset, run quality report, normalize to UnifiedEvent schema, view quality score. Input: CSV file upload (max 8MB, max 100K rows, CSV only), dataset selection parameters. Output: Dataset added to registry, quality report generated, normalized schema written. Expected result: Dataset available for task generation, AI training, and analytics.

4. UPLOAD
Purpose: Upload CSV datasets into the system. What user sees: File upload form with validation status. What buttons do: Select file, upload, cancel. Input: CSV file (MIME type validation, max 8MB, max 100K rows). Output: Uploaded file saved as _uploaded.csv, schema detection, quality report. Expected result: CSV validated and stored; non-CSV or oversized files rejected with error message. File re-encoded to secure format.

5. VALIDATE
Purpose: Validate dataset schema and quality. What user sees: Validation results (missing values, duplicates, invalid numerics, out-of-range, label balance), quality score (0-100 + grade). What buttons do: Run validation, view details, export report (JSON/CSV). Input: Dataset from upload or registry. Output: Validation report with issues list and overall quality score. Expected result: Quality score displayed; issues categorized and fixable via cleaning pipeline. Invalid datasets rejected or marked for cleaning.

6. PROCESS
Purpose: Execute the data processing pipeline. What user sees: Pipeline stages visualized (Validation → Schema Detection → Cleaning → Normalization → Feature Engineering → Storage). What buttons do: Start processing, pause/stop, view progress. Input: Validated dataset. Output: Cleaned, normalized dataset in UnifiedEvent schema, stored in datasets/, quality metrics saved, features extracted for AI. Expected result: Processing completes; dataset ready for AI analysis, scheduling, and analytics. Large datasets processed via chunking/streaming.

7. AI
Purpose: Risk prediction and natural-language explanations. What user sees: Risk level (safe/trained/critical) + confidence score for tasks, explanation buttons, Groq LLM insights (if configured). What buttons do: Analyze task, request explanation, toggle Groq/local AI mode. Input: Task parameters (resource state, attack severity, workload features). Output: Risk prediction from local Random Forest model; if Groq enabled, natural-language explanation from Groq Cloud LLM. Expected result: Risk score and level displayed; Groq insights shown when available, local deterministic fallback otherwise. No crash if Groq unavailable.

8. RISK

Purpose: Security risk assessment and monitoring. What user sees: Real-time risk levels per resource (SAFE/WARNING/HIGH/CRITICAL/QUARANTINED), security timeline, attack events, integrity scores for storage. What buttons do: View audit logs, filter by category/severity, export reports. Input: Digital Twin resource states, attack simulation events, security monitor data. Output: Risk dashboards, alert notifications, exportable reports. Expected result: Current risk state visible; historical trend tracking; audit log exportable for compliance.

9. DIGITAL TWIN

Purpose: Virtual mirror of edge/MEC/cloud resource states for monitoring and decision support. What user sees: Resource cards (CPU, memory, latency, trust, risk level), topology map, real-time state updates, attack impact visualization. What buttons do: Refresh state, apply attack, mitigate attack, quarantine node, reconfigure resources. Input: Task scheduling decisions, attack events, resource utilization updates. Output: Synchronized resource state; risk scores updated; topology reflected in UI. Expected result: Digital Twin stays in sync with scheduler and attack system; resource risk drives GA fitness function and rerouting decisions.

10. GENETIC ALGORITHM

Purpose: Security-aware task scheduling optimization. What user sees: Population parameters (population, generations, mutation/crossover rates), fitness history chart, best assignment results, comparison against baselines (random, greedy, privacy-aware). What buttons do: Run GA, pause, reset, adjust parameters, compare with baselines. Input: Task dataset, resource configuration, GA weights (latency/energy/privacy/security/deadline). Output: Best chromosome (task-to-resource assignment), fitness improvement over generations, success rate, latency, energy, privacy exposure, security risk metrics. Expected result: GA produces optimized schedule; proposed GA should match or beat greedy on deadline success and beat random/greedy on privacy exposure and security risk.

11. SCHEDULER

Purpose: Task-to-resource assignment across Local, MEC, and Cloud. What user sees: Scheduler controls (algorithm selector: random/greedy/privacy-aware/GA), resource topology, latency/energy/privacy/security metrics. What buttons do: Select algorithm, run scheduling, view results, compare baselines. Input: Task dataset, current Digital Twin state, GA parameters (if selected). Output: Task assigned to resources; metrics displayed (success rate, latency, energy, privacy, security risk). Expected result: Scheduling completes; tasks placed; metrics compared against baselines.

12. ATTACK SIMULATION

Purpose: Simulated attack events for testing detection and response. What user sees: Attack type selector (DDoS, Eavesdropping, Tampering, Data Poisoning, Credential Abuse), target node selector, severity slider. What buttons do: Launch attack, mitigate, stop, view timeline. Input: Attack type, target resource, severity (0-1). Output: Attack event recorded in audit log; resource risk increased; Digital Twin updated; security monitor alert triggered; GA reoptimization with new weights; tasks rerouted to healthier resources. Expected result: Safe software simulation only - no real network attacks, no filesystem commands, no database operations. All effects confined to digital twin and virtual storages. Mitigation stops attack, risk decreases, node recovered.

13. STORAGE ATTACK

Purpose: Simulated storage integrity attacks for testing protection and recovery. What user sees: Virtual storages (STORAGE-A: 1000GB encrypted, STORAGE-B: 2000GB encrypted, STORAGE-C: 500GB unencrypted), integrity risk scores, attack events. What buttons do: Simulate attack, isolate, recover, view integrity status. Input: Storage target, attack type. Output: Integrity risk score updated; attack recorded in audit log; affected storage isolated; other storages unaffected; tasks rerouted to available storage. Expected result: Safe storage simulation - virtual only, no real data touched. Integrity restored on recovery; all data remains safe.

14. ANALYTICS

Purpose: Results visualization and performance analysis. What user sees: Charts (success rate, latency, energy, privacy exposure, security risk) comparing scheduler algorithms, experiment campaigns, timeline graphs, risk heatmaps. What buttons do: Filter by dataset/algorithm/date range, export data, generate report. Input: Experiment results data, scheduler metrics, attack timeline. Output: Interactive charts (Chart.js), tabular data, export options (CSV, JSON, PDF). Expected result: Visual comparison of proposed GA vs. baselines;

experiment campaign results stored in experiment_results_2000.csv and /api/export/experiments.

15. EXPERIMENTS

Purpose: Run comparison campaigns between scheduler algorithms with various dataset sizes and AI configurations. What user sees: Campaign configuration (algorithm selection, dataset size, AI enabled/disabled), run status, results summary. What buttons do: Start campaign, pause/stop, view results, export results. Input: Baseline algorithms (random, greedy, privacy-aware, proposed GA), dataset size, AI provider (local/Groq), GA parameters. Output: Experiment results appended to experiment_results_2000.csv; visible in /analytics and via /api/export/experiments; comparison charts. Expected result: Campaign runs all four schedulers on shared workload; results compare success rate, latency, energy, privacy exposure, security risk; proposed GA should outperform on privacy and security; reproducibility via fixed seed.

16. REPORTS

Purpose: Generate documentation and export of results. What user sees: Report type selector (CSV, JSON, PDF), experiment report, system state report, analytics data. What buttons do: Generate CSV, Generate JSON, Generate PDF, download. Input: Selected data range, report type, experiment campaign ID. Output: Downloadable file in selected format. Expected result: CSV export of experiment metrics; JSON export of system state; PDF report with charts and metrics for presentations.

17. SYSTEM HEALTH

Purpose: Monitor service status and component readiness. What user sees: List of monitored modules (backend, dataset, digital_twin, ga, ai, security, attack, storage, analytics), status indicators (online/offline), dataset counts, model readiness, Celery/Redis status. What buttons do: Refresh, reset, navigate to module pages. Input: System state from engine modules and Django database. Output: Health dashboard with green/yellow/red status for each module. Expected result: All critical modules reported; optional modules (Redis, Celery) shown with status; model readiness check; Groq configuration status.

18. SETTINGS

Purpose: Configure application parameters. What user sees: AI provider selection (localfeatures/groq/hybrid), Groq model selection, GA parameters (population, generations, weights), resource configuration, upload limits, rate limiting, dataset seed, debug mode, data source label. What buttons do: Save configuration, reset to defaults, test Groq connection. Input: Environment variables (.env) and Django settings. Output: Configuration applied; restart may required for some changes. Expected result: Settings persist across restarts via .env; Groq key optional; local AI mode works without key; all secrets in .env not in source code.